

# Closing the Pre-Infusion Gap: A Quality Improvement Initiative to Enhance Laboratory Readiness for Outpatient IV Iron Referrals

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## Laboratory completion increased from 69.1% to 89.8%

Laboratory treatment-parameter attainment remained high: 90.8% at baseline and 94.9% after PDSA 2.

### MOTIVATION

At Renown in Northern Nevada, some patients arrived for IV iron infusions without valid iron studies, requiring same-day testing and coordination among nursing staff, pharmacy, and referring clinicians. Lab orders embedded in treatment plans did not reliably ensure readiness before arrival, highlighting the need for earlier verification.

★ Reno  
Nearly 1 in 3 lacked valid iron studies before infusion  
Baseline: 64/207 | Jan–Apr 2026

### AIM

Increase required iron study completion to ≥85% within six months among internal IV iron referrals, supported by verification before scheduling.

### ESTIMATED CHAIR-TIME BURDEN

Baseline period: Jan–Apr 2026  
Assuming 1 additional chair-hour per affected visit

16 additional chair-hours/month  
≈\$18,600/month in charge-based value  
Annualized: 192 hours ≈ \$223,000\*

\*Estimates use a \$1,161.30/hour rate proxy and assume one additional chair-hour per affected visit. Annual projections assume baseline volume and sustained period-specific rates; actual delays, costs, and savings were not measured.

### METHODS

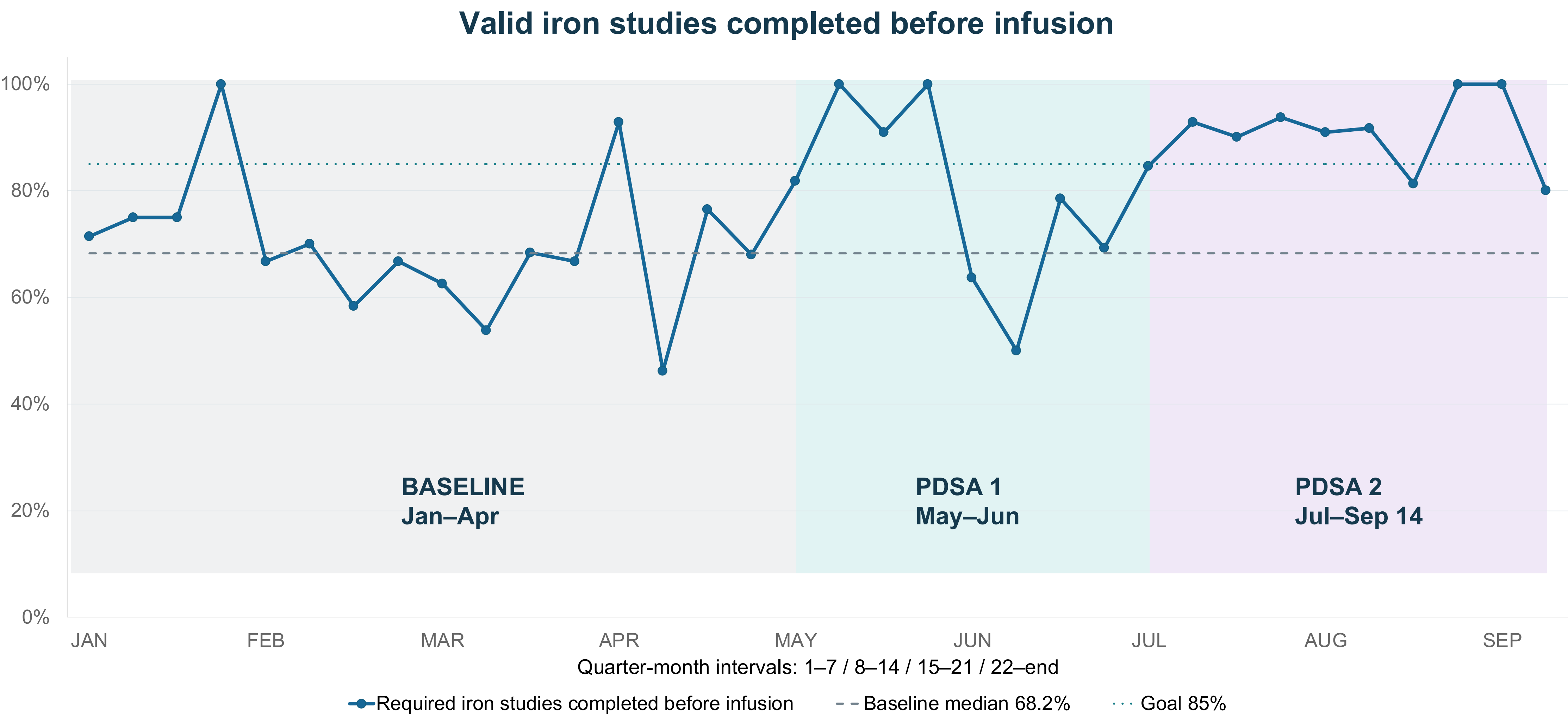
Single-center QI evaluation of 416 internal outpatient IV iron referral records, January 1–September 14, 2026. All records were included regardless of treatment status.

Process measure: Percentage with complete, valid iron studies before infusion.

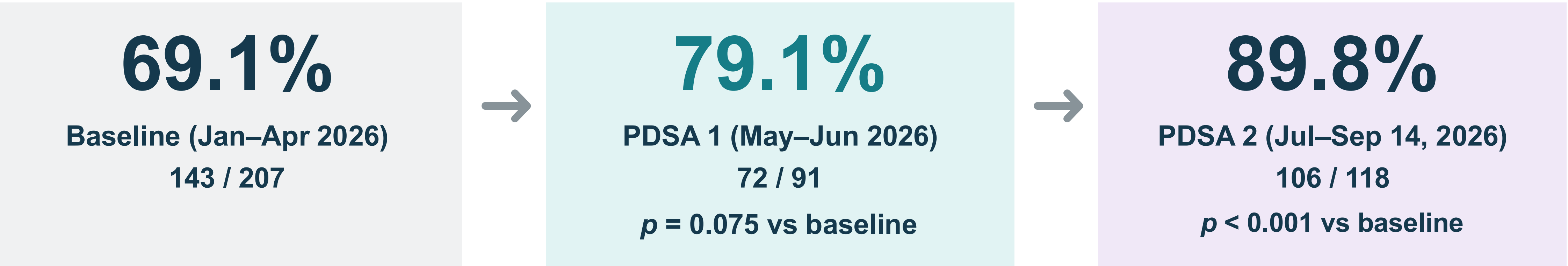
Outcome measure: Percentage meeting laboratory treatment parameters.

Analysis:  
• quarter-month run chart displayed process performance over time using a fixed baseline median.  
• Pearson’s chi-square test compared pooled laboratory readiness rates across the three study periods: baseline, PDSA 1, and PDSA 2, assuming independent records. Outcome rates were summarized descriptively.

### PROCESS MEASURE: LABORATORY READINESS OVER TIME

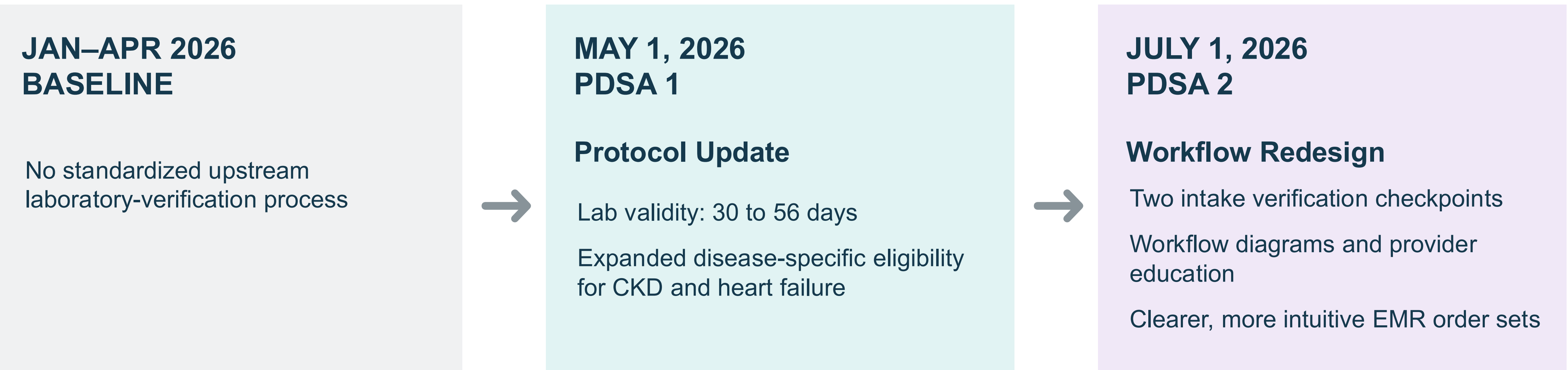


### LABORATORY READINESS BY PERIOD



Overall  $p < 0.001$  across three periods.  
Pairwise p-values compare each intervention period with baseline using Pearson’s chi-square tests with Holm adjustment.

### WHAT WE CHANGED



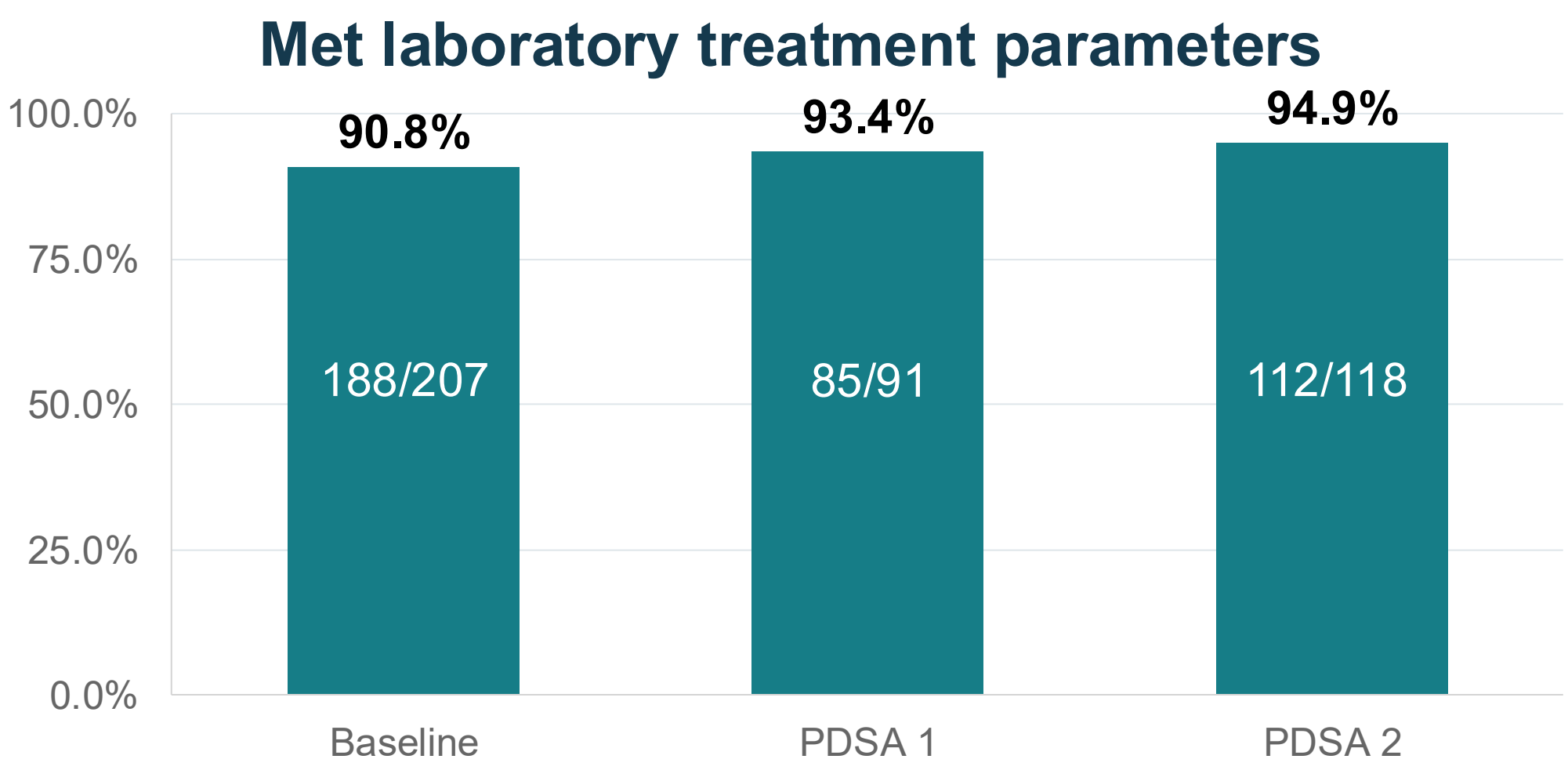
### ACKNOWLEDGMENTS

We thank Renown’s infusion nursing, pharmacy, and intake teams for implementation support, and American Society of Hematology mentors Drs. Ming Lim and Richard Godby for their guidance.

### REFERENCES

Renown Health. *Pricing Transparency*. Accessed September 30, 2026. <https://www.renown.org/Patients-and-Visitors/Billing/Pricing-Transparency>

### OUTCOME MEASURE



Attainment remained high across all periods.

### PATIENT IMPACT (ESTIMATED)

≈129 patient-hours of extra chair time potentially avoided annually\*

Potentially less waiting, fewer treatment disruptions, and less uncertainty for patients and caregivers, including those traveling from rural Northern Nevada.

\*Uses the same chair-time model as the capacity estimates below, assuming baseline volume, one extra hour per affected visit, and sustained improvement. Patient benefits were not directly measured.

### MODELED ANNUAL CAPACITY BURDEN

| Modeled annual burden* | Baseline  | PDSA 1    | PDSA 2   |
|------------------------|-----------|-----------|----------|
| Additional chair-hours | 192       | 130       | 63       |
| Rate-based value       | \$223,000 | \$151,000 | \$73,000 |

\* Annualized at baseline volume (621 visits/year), assuming one additional chair-hour per visit lacking valid studies and \$1,161.30/hour. Projections assume sustained period-specific rates; not measured delays or financial savings.

### LIMITATIONS

Single-center, uncontrolled evaluation with limited follow-up. Changes in laboratory validity and eligibility criteria may affect comparisons across periods. Individual intervention effects cannot be isolated. Chair-time and financial estimates are modeled, not directly measured.

### CONCLUSION

Protocol and workflow changes were associated with improved laboratory readiness, exceeding the 85% target overall after PDSA 2. Laboratory treatment-parameter attainment remained high. Intermittent below-target intervals highlight the need for continued monitoring and refinement.

### NEXT STEPS

Monitor sustainability and adapt the workflow for an external-referral pilot.

